

Publications

Books

- [1] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, *Design Automation for Field-coupled Nanotechnologies*. Springer, 2022. DOI: 10.1007/978-3-030-89952-3

Journal Articles

- [2] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, “RTL-to-Atoms Synthesis of a Machine Learning Accelerator on Atomic-Scale Computers,” *IEEE Transactions on Nanotechnology (TNANO)*, 2026. DOI: 10.1109/TNANO.2026.3689032
- [3] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “QuickCell: Fast Automatic Design of Standard Cells for Silicon Dangling Bond Logic,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3605537
- [4] S. Hofmann, **M. Walter**, R. Wille, “Efficient and Scalable Post-Layout Optimization for Field-coupled Nanotechnologies,” *IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. DOI: 10.1109/TCAD.2025.3549354
- [5] S. Hofmann, **M. Walter**, R. Wille, “Graph-Oriented Layout Design for Field-coupled Nanocomputing via Parallel Multi-Objective Search Space Exploration,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2025. DOI: 10.1109/TCSI.2025.3614709
- [6] S. S. H. Ng, J. Croshaw, **M. Walter**, R. Wille, R. Wolkow, K. Walus, “Simulating Charged Defects in Silicon Dangling Bond Logic Systems to Evaluate Logic Robustness,” *IEEE Transactions on Nanotechnology (TNANO)*, 2024. DOI: 10.1109/TNANO.2024.3372946
- [7] M. D. Vieira, S. S. H. Ng, **M. Walter**, R. Wille, K. Walus, R. S. Ferreira, O. P. Vilela Neto, J. A. M. Nacif, “Three-Input NPN Class Gate Library for Atomic Silicon Quantum Dots,” *IEEE Design & Test*, 2022. DOI: 10.1109/MDAT.2022.3189814
- [8] F. Sill Torres, P. A. Silva, G. Fontes, **M. Walter**, J. A. M. Nacif, R. S. Ferreira, O. P. V. Neto, J. F. Chaves, R. Wille, P. Niemann, D. Große, R. Drechsler, “On the Impact of the Synchronization Constraint and Interconnections in Quantum-dot Cellular Automata,” *Microprocessors and Microsystems (MICPRO)*, 2020. DOI: 10.1016/j.micpro.2020.103109
- [9] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, “Placement and Routing for Tile-based Field-coupled Nanocomputing Circuits is NP-complete (Research Note),” *Journal on Emerging Technologies in Computing Systems (JETC)*, 2019. DOI: 10.1145/3312661

Conference Proceedings

- [10] L. Burgholzer, **M. Walter**, P. Hopf, A. Caride-Tabarés Sánchez, T. Mattsson, B. Hoffmann, N. Färkkilä, D. Bulmash, R. Wille, E. Mansfield, “Practical HPCQC Integration with QDMI: A Real-Hardware Case Study with IQM Systems,” in *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, 2026.
- [11] J. Drewniok, **M. Walter**, R. Wille, “Operational Domain Sketching for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [12] M. Feldmeier, **M. Walter**, R. Wille, “A Framework for Post-Mapping Feedback-Guided Selective Logic Transformations,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [13] I. V. Gardner, **M. Walter**, R. Wille, M. Cochez, “AIG2PT: A Generative Pre-trained Transformer for Unconditional And-Inverter Graph Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [14] B. Hien, **M. Walter**, R. Wille, “Minimizing Area and Delay in Planar Physical Design for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [15] B. Hien, **M. Walter**, R. Wille, “Toward Compact Planar Circuits for Atomic-Scale Computing: A Hybrid Crossing Elimination Flow,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.

- [16] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Holistic Physical Design for Silicon Atomic Logic: Cramming More Components onto Clock Zones,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [17] W. Lambooy, J. Drewniok, **M. Walter**, R. Wille, “Mastering the Exponential Complexity of Exact Physical Simulation of Silicon Dangling Bonds,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2026. DOI: 10.1109/ASP-DAC66049.2026.11420323
- [18] S. S. H. Ng, **M. Walter**, S. Hofmann, J. Drewniok, R. Wille, K. Walus, “Design and Emulation Methodology for Atomic-Scale Systolic Arrays: An LLM Accelerator Case Study in Silicon DB Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [19] S. S. H. Ng, **M. Walter**, M. Yuan, J. Drewniok, R. H. Foote, R. Lupoiu, J. A. Fan, R. Wolkow, R. Wille, K. Walus, “Bit-PlanarNet: Learning-to-Layout Planar Gate Networks for Emerging Devices,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2026.
- [20] **M. Walter**, M. Feldmeier, R. Wille, “Exact Synthesis with Optimal Switching Activity,” in *Design, Automation and Test in Europe (DATE)*, 2026. DOI: 10.23919/DATE69613.2026.11539139
- [21] **M. Walter**, I. V. Gardner, R. Wille, “aigverse: A Unified Infrastructure for Machine Learning-Driven Logic Synthesis,” in *International Workshop on Logic Synthesis (IWLS)*, 2026.
- [22] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Towards Fast Automatic Design of Silicon Dangling Bond Logic,” in *Design, Automation and Test in Europe (DATE)*, 2025. DOI: 10.23919/DATE64628.2025.10992885
- [23] J. Drewniok, **M. Walter**, R. Wille, “QuickTrace: An Efficient Contour Tracing Algorithm for Defect Robustness Simulation of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025. DOI: 10.1109/ISCAS56072.2025.11044082
- [24] I. V. Gardner, **M. Walter**, Y. Miyasaka, R. Wille, M. Cochez, “Bias by Design: Diversity Quantification to Mitigate Structural Bias Effects in AIG Logic Optimization,” in *Design, Automation and Test in Europe (DATE)*, 2025. DOI: 10.23919/DATE64628.2025.10993009
- [25] B. Hien, D. Quinci, Y. Ardesi, G. Beretta, F. Ravera, **M. Walter**, R. Wille, “Bridging the Gap Between Molecular FCN and Design Automation with SIM(7)-MolPDK: A Physically Simulated Standard-Cell Library,” in *IEEE Latin American Conference on Nanotechnology (IEEE LANANO)*, 2025. DOI: 10.1109/LANANO68466.2025.11512469
- [26] B. Hien, **M. Walter**, A. Tempia Calvino, A. Mishchenko, R. Wille, “Ashenhurst-Curtis Decomposition Using Don’t Cares,” in *International Workshop on Logic Synthesis (IWLS)*, 2025.
- [27] B. Hien, **M. Walter**, S. Hofmann, R. Wille, “A Fully Planar Approach to Field-coupled Nanocomputing: Scalable Placement and Routing Without Wire Crossings,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025. DOI: 10.1109/NANO63165.2025.11113822
- [28] S. Hofmann, J. Drewniok, **M. Walter**, R. Wille, “MNT Designer: A Comprehensive Design Tool for Field-coupled Nanocomputing,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025. DOI: 10.1109/NANO63165.2025.11113823
- [29] S. Hofmann, **M. Walter**, R. Wille, “Late Breaking Results: Physical Co-Design for Field-coupled Nanocomputing,” in *Design, Automation and Test in Europe (DATE)*, 2025. DOI: 10.23919/DATE64628.2025.10993138
- [30] S. Hofmann, **M. Walter**, R. Wille, “Physical Design for Field-coupled Nanocomputing with Discretionary Cost Objectives,” in *IEEE Latin American Symposium on Circuits and Systems (LASCAS)*, Best Paper Award Candidate, 2025.
- [31] S. S. H. Ng, **M. Walter**, J. Drewniok, S. Hofmann, R. Wille, K. Walus, “Building a Machine Learning Accelerator with Silicon Dangling Bonds: From Verilog to Quantum Dot Layout,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, Best Student Paper Award, 2025. DOI: 10.1109/NANO63165.2025.11113747
- [32] **M. Walter**, J. Drewniok, R. Wille, “Live Demonstration: An Application for Layout Resilience Analysis of Silicon Dangling Bond Logic,” in *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025. DOI: 10.1109/ISCAS56072.2025.11043498
- [33] **M. Walter**, J. Drewniok, R. Wille, “The Operational Domain Explorer: A Comprehensive Framework to Unveil the Thermal Landscape of Silicon Dangling Bond Logic Beyond Conventional Operability,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2025. DOI: 10.1109/NANO63165.2025.11113672
- [34] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “On-the-fly Defect-Aware Design of Circuits based on Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628962
- [35] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, “Unifying Figures of Merit: A Versatile Cost Function for Silicon Dangling Bond Logic,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628671

- [36] J. Drewniok, **M. Walter**, R. Wille, "The Need for Speed: Efficient Exact Simulation of Silicon Dangling Bond Logic," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2024. DOI: 10.1109/ASP-DAC58780.2024.10473946
- [37] B. Hien, **M. Walter**, V. M. Santen, F. Klemme, S. S. Parihar, G. Pahwa, Y. S. Chauhan, H. Amrouch, R. Wille, "Technology Mapping for Cryogenic CMOS Circuits," in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2024. DOI: 10.1109/ISVLSI61997.2024.00057
- [38] B. Hien, **M. Walter**, R. Wille, "Reducing Wire Crossings in Field-Coupled Nanotechnologies," in *International Workshop on Logic Synthesis (IWLS)*, 2024.
- [39] B. Hien, **M. Walter**, R. Wille, "Reducing Wire Crossings in Field-Coupled Nanotechnologies," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628717
- [40] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, "Thinking Outside the Clock: Physical Design for Field-coupled Nanocomputing with Deep Reinforcement Learning," in *International Symposium on Quality Electronic Design (ISQED)*, 2024. DOI: 10.1109/ISQED60706.2024.10528711
- [41] S. Hofmann, **M. Walter**, R. Wille, "A* is Born: Efficient and Scalable Physical Design for Field-coupled Nanocomputing," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628808
- [42] S. Hofmann, **M. Walter**, R. Wille, "Late Breaking Results: Wiring Reduction for Field-coupled Nanotechnologies," in *Design Automation Conference (DAC)*, 2024. DOI: 10.1145/3649329.3663491
- [43] S. Hofmann, **M. Walter**, R. Wille, "MNT Bench: Benchmarking Software and Layout Libraries for Field-coupled Nanocomputing," in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546596
- [44] D. Marakkalage, **M. Walter**, S.-Y. Lee, R. Wille, G. De Micheli, "Technology Mapping for Beyond-CMOS Circuitry with Unconventional Cost Functions," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, Best Presentation Award, 2024. DOI: 10.1109/NANO61778.2024.10628909
- [45] S. S. H. Ng, J. Drewniok, **M. Walter**, J. Retallick, R. Wille, K. Walus, "Unlocking Flexible Silicon Dangling Bond Logic Designs on Alternative Silicon Orientations," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628802
- [46] **M. Walter**, J. Croshaw, S. S. H. Ng, K. Walus, R. Wolkow, R. Wille, "Towards Atomic Defect-Aware Physical Design of Silicon Dangling Bond Logic on the H-Si(100)-2x1 Surface," in *Design, Automation and Test in Europe (DATE)*, 2024. DOI: 10.23919/DATE58400.2024.10546683
- [47] **M. Walter**, J. Drewniok, S. Hofmann, B. Hien, R. Wille, "The Munich Nanotech Toolkit (MNT)," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628747
- [48] **M. Walter**, J. Drewniok, R. Wille, "Ending the Tyranny of the Clock: SAT-based Clock Number Assignment for Field-coupled Nanotechnologies," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2024. DOI: 10.1109/NANO61778.2024.10628908
- [49] J. Drewniok, **M. Walter**, S. S. H. Ng, K. Walus, R. Wille, "QuickSim: Efficient and Accurate Physical Simulation of Silicon Dangling Bond Logic," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231266
- [50] J. Drewniok, **M. Walter**, R. Wille, "Minimal Design of SiDB Gates: An Optimal Basis for Circuits Based on Silicon Dangling Bonds," in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633241
- [51] J. Drewniok, **M. Walter**, R. Wille, "Temperature Behavior of Silicon Dangling Bond Logic," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231259
- [52] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, "Late Breaking Results From Hybrid Design Automation for Field-coupled Nanotechnologies," in *Design Automation Conference (DAC)*, 2023. DOI: 10.1109/DAC56929.2023.10247933
- [53] S. Hofmann, **M. Walter**, L. Servadei, R. Wille, "Thinking Outside the Clock: Physical Design for Field-coupled Nanocomputing with Deep Reinforcement Learning," in *International Workshop on Logic Synthesis (IWLS)*, 2023.
- [54] S. Hofmann, **M. Walter**, R. Wille, "Post-Layout Optimization for Field-coupled Nanotechnologies," in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. DOI: 10.1145/3611315.3633247
- [55] S. Hofmann, **M. Walter**, R. Wille, "Scalable Physical Design for Silicon Dangling Bond Logic: How a 45°~Turn Prevents the Reinvention of the Wheel," in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2023. DOI: 10.1109/NANO58406.2023.10231278

- [56] D. Marakkalage, **M. Walter**, S.-Y. Lee, R. Wille, G. De Micheli, “Technology Mapping for Beyond-CMOS Circuitry with Unconventional Cost Functions,” in *International Workshop on Logic Synthesis (IWLS)*, Best Paper Award Candidate, 2023.
- [57] V. M. Santen, **M. Walter**, F. Klemme, S. S. Parihar, G. Pahwa, Y. Chauhan, R. Wille, H. Amrouch, “Design Automation for Cryogenic CMOS Circuits,” in *International Workshop on Logic Synthesis (IWLS)*, 2023.
- [58] V. M. Santen, **M. Walter**, F. Klemme, S. S. Parihar, G. Pahwa, Y. Chauhan, R. Wille, H. Amrouch, “Design Automation for Cryogenic CMOS Circuits,” in *Design Automation Conference (DAC)*, 2023. doi: 10.1109/DAC56929.2023.10247824
- [59] **M. Walter**, J. Drewniok, S. S. H. Ng, K. Walus, R. Wille, “Reducing the Complexity of Operational Domain Computation in Silicon Dangling Bond Logic,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2023. doi: 10.1145/3611315.3633246
- [60] **M. Walter**, B. Hien, R. Wille, “Versatile Signal Distribution Networks for Scalable Placement and Routing of Field-coupled Nanocomputing Technologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2023. doi: 10.1109/ISVLSI59464.2023.10238604
- [61] W. Lambooy, **M. Walter**, R. Wille, “Exploiting the Third Dimension: Stackable Quantum-dot Cellular Automata,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2022. doi: 10.1145/3565478.3572529
- [62] **M. Walter**, S. S. Hang Ng, K. Walus, R. Wille, “Hexagons are the Bestagons: Design Automation for Silicon Dangling Bond Logic,” in *Design Automation Conference (DAC)*, 2022. doi: 10.1145/3489517.3530525
- [63] **M. Walter**, R. Wille, “Efficient Multi-Path Signal Routing for Field-coupled Nanotechnologies,” in *International Symposium on Nanoscale Architectures (NANOARCH)*, 2022. doi: 10.1145/3565478.3572539
- [64] **M. Walter**, W. Haaswijk, R. Wille, F. Sill Torres, R. Drechsler, “One-Pass Synthesis for Field-coupled Nanocomputing Technologies,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Best Paper Award Candidate, 2021. doi: 10.1145/3394885.3431607
- [65] U. Garlando, **M. Walter**, R. Wille, F. Riente, F. Sill Torres, R. Drechsler, “ToPoliNano and fiction: Design Tools for Field-coupled Nanocomputing,” in *Euromicro Conference on Digital System Design (DSD)*, 2020. doi: 10.1109/DSD51259.2020.00071
- [66] T. Meywerk, **M. Walter**, D. Große, R. Drechsler, “Clustering-Guided SMT(LRA) Learning,” in *International Conference on Integrated Formal Methods (IFM)*, 2020. doi: 10.1007/978-3-030-63461-2_3
- [67] T. Meywerk, **M. Walter**, V. Herdt, J. Kleinekathöfer, D. Große, R. Drechsler, “Verifying Safety Properties of Robotic Plans Operating in Real-World Environments via Logic-Based Environment Modeling,” in *International Symposium on Leveraging Applications of Formal Methods (ISoLA)*, 2020. doi: 10.1007/978-3-030-61470-6_20
- [68] **M. Walter**, R. Drechsler, “Design Automation for Field-coupled Nanotechnologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Best Student Forum Paper Award, 2020, pp. 176–181. doi: 10.1109/ISVLSI49217.2020.00040
- [69] **M. Walter**, W. Haaswijk, R. Wille, F. Sill Torres, R. Drechsler, “SAT-based Exact Physical Design for Field-coupled Nanocomputing Technologies,” in *International Workshop on Logic Synthesis (IWLS)*, 2020.
- [70] **M. Walter**, R. Wille, F. Sill Torres, R. Drechsler, “Bail on Balancing: An Alternative Approach to the Physical Design of Field-coupled Nanocomputing Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Best Paper Award Candidate, 2020. doi: 10.1109/ISVLSI49217.2020.00022
- [71] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “Verification for Field-coupled Nanocomputing Circuits,” in *Design Automation Conference (DAC)*, 2020. doi: 10.5555/3437539.3437614
- [72] T. Meywerk, **M. Walter**, V. Herdt, D. Große, R. Drechsler, “Towards Formal Verification of Plans for Cognition-enabled Autonomous Robotic Agents,” in *Euromicro Conference on Digital System Design (DSD)*, 2019. doi: 10.1109/DSD.2019.00029
- [73] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “fiction: An Open Source Framework for the Design of Field-coupled Nanocomputing Circuits,” in *International Workshop on Logic Synthesis (IWLS)*, 2019.
- [74] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “fiction: Electronic Design Automation for Field-coupled Nanocomputing Circuits,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Best Research Demo Award, 2019.
- [75] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “Scalable Design for Field-coupled Nanocomputing Circuits,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2019. doi: 10.1145/3287624.3287667
- [76] R. Wille, **M. Walter**, F. Sill Torres, D. Große, R. Drechsler, “Ignore Clocking Constraints: An Alternative Physical Design Methodology for Field-coupled Nanotechnologies,” in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2019. doi: 10.1109/ISVLSI.2019.00121

- [77] F. Sill Torres, **M. Walter**, R. Wille, D. Große, R. Drechsler, “Synchronization of Clocked Field-Coupled Circuits,” in *IEEE International Conference on Nanotechnology (IEEE Nano)*, 2018. doi: 10.1109/NANO.2018.8626294
- [78] F. Sill Torres, R. Wille, **M. Walter**, P. Niemann, D. Große, R. Drechsler, “Evaluating the Impact of Interconnections in Quantum-Dot Cellular Automata,” in *Euromicro Conference on Digital System Design (DSD)*, 2018. doi: 10.1109/DSD.2018.00110
- [79] **M. Walter**, R. Wille, D. Große, F. Sill Torres, R. Drechsler, “An Exact Method for Design Exploration of Quantum-dot Cellular Automata,” in *Design, Automation and Test in Europe (DATE)*, 2018. doi: 10.23919/DATE.2018.8342060
- [80] R. Wille, O. Keszocze, S. Hillmich, **M. Walter**, A. Garcia-Ortiz, “Synthesis of Approximate Coders for On-chip Interconnects Using Reversible Logic,” in *Design, Automation and Test in Europe (DATE)*, 2016. doi: 10.1109/DATE.2016.7459481
- [81] R. Wille, O. Keszocze, **M. Walter**, P. Rohrs, A. Chattopadhyay, R. Drechsler, “Look-ahead Schemes for Nearest Neighbor Optimization of 1D and 2D Quantum Circuits,” in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2016. doi: 10.1109/ASPDAC.2016.7428026

Preprints

- [82] **M. Walter**, J. Croshaw, S. S. H. Ng, K. Walus, R. Wolkow, R. Wille, “Atomic Defect-Aware Physical Design of Silicon Dangling Bond Logic on the H-Si(100)2x1 Surface,” 2023. eprint: 2311.12042.
- [83] **M. Walter**, R. Wille, F. Sill Torres, D. Große, R. Drechsler, “fiction: An Open Source Framework for the Design of Field-coupled Nanocomputing Circuits,” 2019. eprint: 1905.02477.